

Pharmacist Involvement in Specialty Medication Discontinuations and Dose Changes

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Purpose

Data describing HSSP pharmacists' role and financial implications in intervening to discontinue or switch ineffective or unsafe treatments is needed. The purpose of this study is to evaluate the health-system specialty pharmacy (HSSP) pharmacists' involvement in specialty medication discontinuations, switches, and dose changes and the resulting cost avoidance.

Study Design and Population

Single-center, retrospective cohort analysis of data collected from an electronic medical record and specialty pharmacy management system.

Patients were included if they had an intervention where the HSSP pharmacist indicated the discontinuation, dose change or switch resulted from the pharmacist's recommendation.

Methods

Defining medication outcomes

Discontinuation: Medication stopped with no new specialty medication started within 30 days
Switch: Medication stopped and patient started a different specialty medication
Dose change: Medication stayed the same, but with a different dose

Scoring outcome probability

A 3-investigator panel categorized each intervention on the probability (Pr) that the medication outcome occurred due to the pharmacist:
from absolutely not [Pr=0] to likely [Pr=0.75]

Calculating cost avoidance

Medication costs calculated using:

- Wholesale acquisition cost (WAC)
- Average wholesale price (AWP)
- AWP minus 20%

Cost avoidance calculation:
1 fill of medication costs X the assigned Pr of the medication outcome

Figure 1. Study Attrition

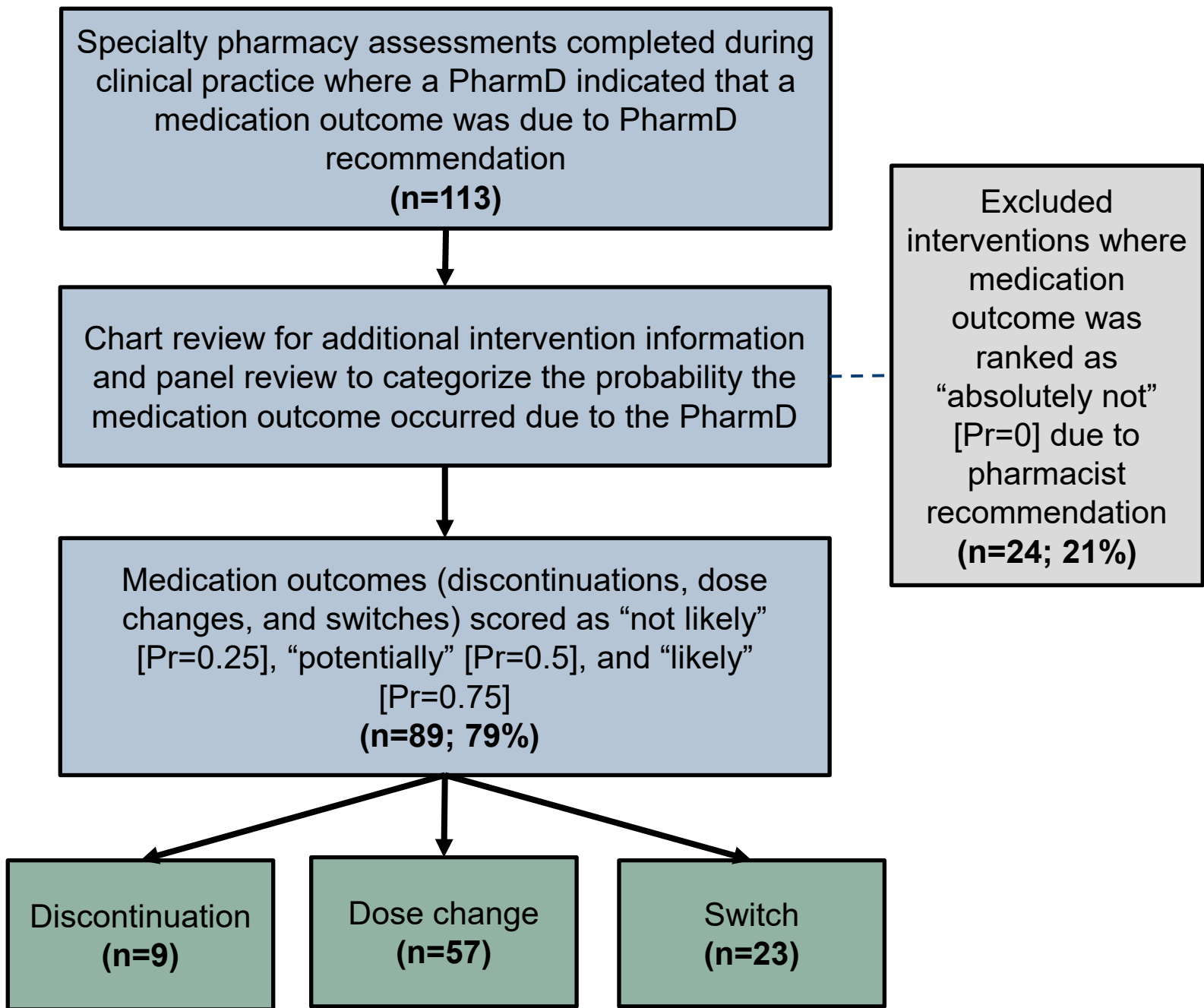


Table 1. Patient Demographics (n=91)

Characteristics	n (%)
Age (at time of intervention), median (IQR)	53 (27 – 65)
Gender	
Female	55 (60)
Male	36 (40)
Race, White	76 (84)
Pharmacy Insurance	
Medicare	38 (46)
Commercial	23 (28)
Medicaid	17 (21)
Clinic	
Neurology	23 (25)
Rheumatology	27 (30)

Figure 3. Intervention Reasons

Reason for Intervention	Discontinuation	Dose change	Switch
Weight-based dose adjustment	0	5	0
Non-adherence	3	1	1
Dose titration	0	5	0
Other	1	6	3
Administration issues	0	12	0
Side effects	4	7	10
Medication no longer working	1	21	9

Medication outcomes that were at least somewhat likely due to the pharmacist intervention that resulted in cost avoidance are presented.

HIGHLIGHTS

- Pharmacists avoided \$129,705 - \$162,133 in direct costs through interventions.
- Health system specialty pharmacy pharmacists intervene to support safe and effective specialty medication use and avoid unnecessary costs.
- These results can be used to rationalize the employment and support for pharmacists in the specialty pharmacy setting beyond filling prescriptions.

Results

Figure 2. Costs Per Fill Avoided

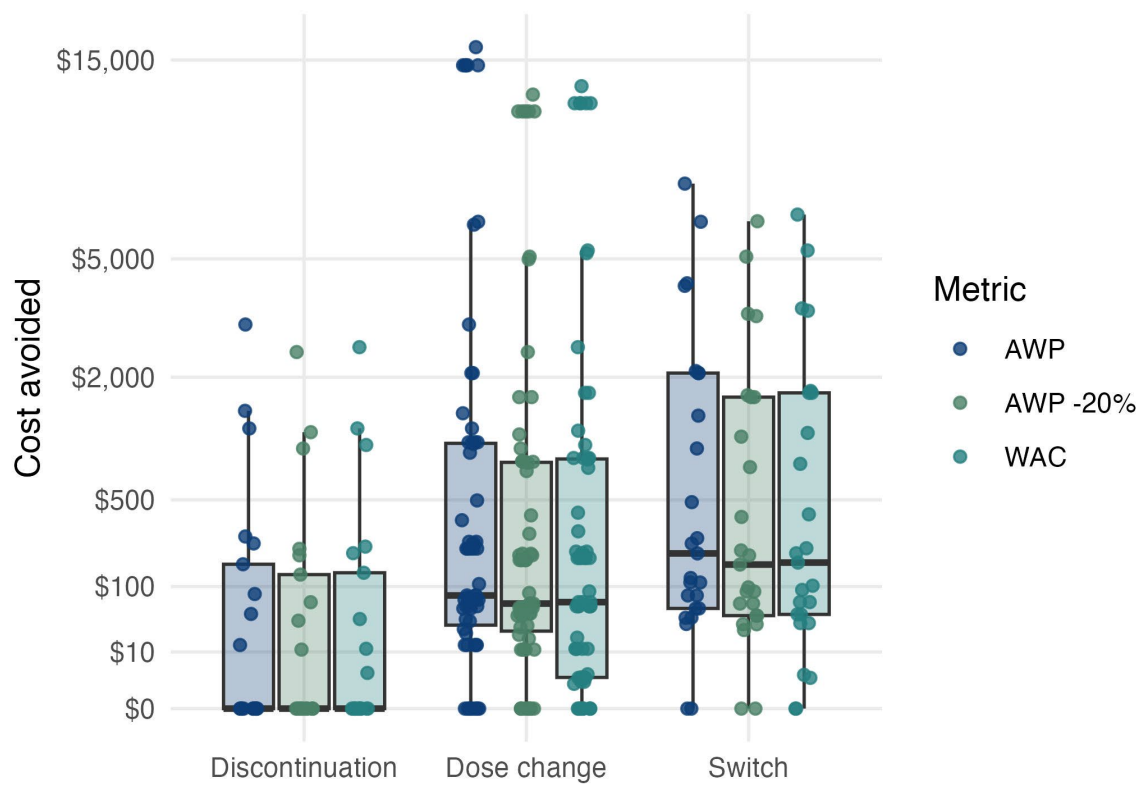
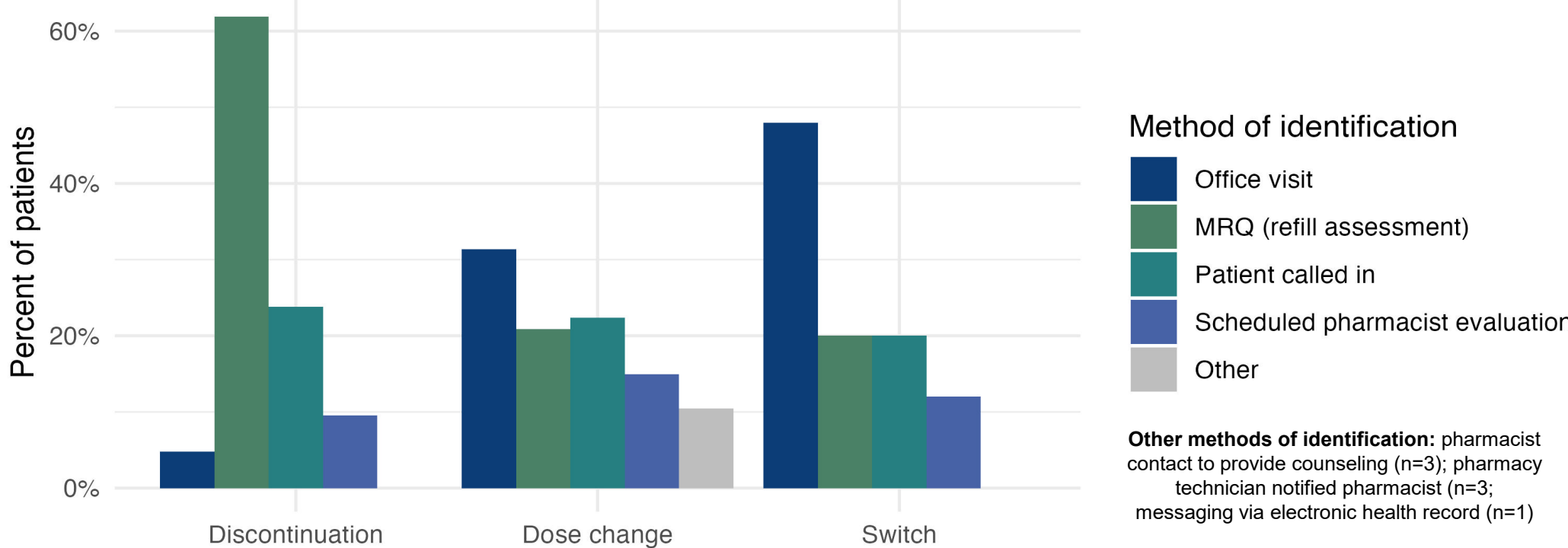


Table 2. Total Cost Avoidance

Cost Avoidance by Medication ^a	
Medication ^b	Cost Avoidance
Emicizumab-kxwh	\$73,187
Ustekinumab	\$23,865
Etanercept	\$19,033
Adalimumab	\$17,653
Secukinumab	\$10,386
Cost Avoidance by Clinic ^a	
Clinic	Cost Avoidance
Hematology	\$73,773
Rheumatology	\$45,330
Peds GI/IBD	\$15,910
Dermatology	\$8,309
Adult GI/IBD	\$7,980
Total Monthly Estimated Cost Range	
Discontinuations	\$5,290 - \$6,613
Dose Changes	\$97,824 - \$122,281
Switches	\$26,591 - \$33,239

^aCost avoidance reported for one month AWP
^bRepresents top 5 medications with cost avoidance

Figure 4. Method of Pharmacist Identification of Intervention Need



Method of identification

- Office visit
- MRQ (refill assessment)
- Patient called in
- Scheduled pharmacist evaluation
- Other

Other methods of identification: pharmacist contact to provide counseling (n=3); pharmacy technician notified pharmacist (n=3); messaging via electronic health record (n=1)